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TERM END EXAMINATION
SUMMER - 2023

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE : CM5460 – COMPUTER GRAPHICS

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

Q.1.(A) Attempt any TWO.

- a) List any 4 graphics file formats. Explain any two.
- b) Explain Liquid Crystal Display with diagram.
- c) Draw and explain Cartesian coordinate system.

08

Q.2.(A) Attempt any ONE.

- a) Give any 2 advantages and disadvantages of DDA line drawing algorithm.
- b) Define polygon. Name various types of polygon.

02

(B) Attempt any THREE.

- a) Define polygon filling. Write the flood fill procedure of polygon.
- b) Rasterize the line from (6,6) to (12,9) using Bresenham's algorithm.
- c) Write steps of the Midpoint circle drawing algorithm.
- d) Consider a point inside the polygon and another point outside the polygon, draw and explain Inside Outside Test of polygon for both the points.

12

Q.3.(A) Attempt any ONE.

- a) Define 3D transformation.
- b) Define shearing.

02

(B) Attempt any THREE.

- a) Explain and write the 2D transformation matrix for : i) Rotation ii) Scaling
- b) Draw and explain 3D transformation for scaling with respect to origine.
- c) Derive 2D transformation matrix for rotation about an arbitrary point.
- d) Find transformation matrix for :

12

i) Translate a polygon with coordinates A (2,5), B (7,10), C (10,2) by 3 units in X direction and 2 units in Y direction.

ii) A point is rotated counter clock wise by an angle of 45° . Find the rotation matrix for the point.



Q.4.(A) Attempt any ONE.

02

- a) Compare viewport and Window.
- b) Define polygon clipping.

(B) Attempt any THREE.

12

- a) Draw and explain viewing transformation.
- b) Write Sutherland and Hodgman polygon clipping algorithm.
- c) Find the normalization transformation window to viewport, with window, lower left corner at (1,1) and upper right corner at (2,7) onto a viewport with lower left corner at (0,0) and upper right corner at ($\frac{1}{2}$, $\frac{1}{2}$).
- d) Write steps of Cohen Sutherland line clipping algorithm.

Q.5.(A) Attempt any TWO.

04

- a) State any 4 problems in True Curve Generation approach.
- b) Draw the second approximation of Koch Curve.
- c) Write any 4 features of Hilbert's curve.

(B) Attempt any TWO.

08

- a) Explain Fractal Lines and Fractal Surfaces with suitable examples.
- b) Write DDA algorithm for circular Arc generation.
- c) Write steps to generate Bezier Curve.

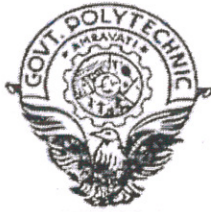
Q.6. Attempt any TWO.

08

- a) State and explain any 4 Hazards of graphics standards.
- b) Write and explain any 4 basic operations of Open GL in brief.
- c) Write any 2 features of GUI and any 2 principles for good GUI design.

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TERM END EXAMINATION
SUMMER - 2023

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE : CM5461-PROGRAMMING WITH PYTHON

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

Q.1.(A) Attempt any TWO.

04

- a) Define input and output in python.
- b) Recall the terms with examples : i) variable ii) keywords
- c) Enlist any four features of python.

(B) Attempt any ONE.

04

- a) Describe the term indentation in python.
- b) Demonstrate the variable declaration & initialization with example in python.

Q.2.(A) Attempt any ONE.

02

- a) Define dictionary in python.
- b) Write the syntax of if-else statement in python.

(B) Attempt any TWO.

08

- a) Write a python program to print the pattern.

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*  
* *  
* * *  
* * * *
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- b) Describe two Membership and two logical operations in python with appropriate examples.
- c) Demonstrate use of break and continue statement with an example.

Q.3.(A) Attempt any ONE.

04

- a) Describe tuple and write steps to create, access and delete tuple items.
- b) Describe dictionary and write steps to access and traverse dictionary items.



(B) Attempt any **TWO**.

08

- Describe Indexing and slicing operations on list in python.
- Write a python program that count number of occurrence of a letter in a string using dictionaries for example string is "MISSISSIPPI" store count of every letter with the letter in dictionary.
- Demonstrate the various operations perform on tuple with the help of example.

Q.4.(A) Attempt any **ONE**.

02

- Define local and global variables.
- Write syntax of function and state use of function.

(B) Attempt any **THREE**.

12

- Write a python program to calculate factorial of given number using function.
- Describe index error exception handling by giving suitable example.
- Describe python package by giving suitable example.
- Write a python function to print Armstrong number between 1 to 1000 using function.

Q.5.(A) Attempt any **TWO**.

04

- Write the different modes of files (a) r+ (b) wb+.
- Write down the steps to open and close file in python.
- Enlist the file object attributes of a file.

(B) Attempt any **TWO**.

08

- Describe different file object attributes and their significance.
- Write a python program to write following user input details to the file :
i) Name ii) email iii) Mobile iv) address
- Write a python program to read a file and count the number of character number of digits present in the file.

Q.6.(A) Attempt any **ONE**.

02

- Define class in python.
- List any two features of object oriented programming in python.

(B) Attempt any **THREE**.

12

- Describe Data Abstraction in python with an example.
- Write a python program for following create a temperature class. Make two methods :
i) Convert Fahrenheit - It will take Celsius and will print it into Fahrenheit.
ii) Convert Celsius - It will take Fahrenheit and will convert it into Celsius.
- Write python program to design student class with following instance variable :
i) roll_no ii) Name and methods to collect roll_no and name.
- Describe the term inheritance in python. Demonstrate the single inheritance.

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TERM END EXAMINATION
SUMMER - 2023

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE : CM5465-CLOUD COMPUTING

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

Q.1.(A) Attempt any ONE.

a) State private cloud.

02

b) Define Paas service model.

(B) Attempt any TWO.

a) Differentiate between cloud service models (any four points).

08

b) Describe characteristics of cloud computing.

c) Explain the architecture of cloud computing.

Q.2.(A) Attempt any TWO.

a) Enlist any four advantages of virtualization.

04

b) Define KVM.

c) List advantages and disadvantages of consolidation (any two points).

(B) Attempt any ONE.

a) Describe VM migration.

04

b) Describe Desktop and application virtualization.

Q.3.(A) Attempt any TWO.

a) Define virtual storage area network.

04

b) Explain virtual provisioning.

c) State need of GFS (Google File System).

(B) Attempt any THREE.

a) Draw and explain architecture of HDFS.

12

b) Describe File level storage.

c) Compare GFS and HDFS.

d) Describe the architecture of virtual data centre (VDC).



Q.4.(A) Attempt any TWO.

04

- a) List the types of SLA.
- b) Define federated cloud.
- c) Write objective of service catalog management. (any four points)

(B) Attempt any THREE.

12

- a) Explain life cycle of SLA.
- b) Describe cloud portal and its function.
- c) Explain Ab-initio Resource optimization.
- d) Explain an architecture of federated cloud computing.

Q.5.(A) Attempt any ONE.

02

- a) Define following term regarding risk. i) Data loss ii) DDOS attack
- b) State the advantages of cloud computing security.

(B) Attempt any TWO.

08

- a) Explain content level security.
- b) Write steps to implement cloud based identity and access management.
- c) Describe technical risk in detail.

Q.6.(A) Attempt any ONE.

02

- a) Define Ubiquitous computing.
- b) List the benefits of smart grid.

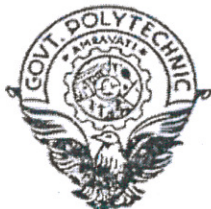
(B) Attempt any TWO.

08

- a) Describe different enabling technologies with Internet of Things (IOT).
- b) Describe the features of Microsoft Azure.
- c) Differentiate between open Nebula and Open stack.

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TERM END EXAMINATION
SUMMER - 2023

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE : CM5467-COMPUTER SECURITY

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

Q.1.(A) Attempt any TWO.

08

- a) Describe in brief CIA model.
- b) What is multipartite and microvirus.
- c) Explain in brief hotfix, patch and service pack.

Q.2.(A) Attempt any TWO.

04

- a) List any four password policies.
- b) Classify authentication methods.
- c) Explain piggybacking with example.

(B) Attempt any TWO.

08

- a) Explain working of biometric system with neat diagram.
- b) What is access control? Explain mandatory access control in brief.
- c) Explain in brief password cracking types.

Q.3. Attempt any THREE.

12

- a) Differentiate between symmetric and Asymmetric key cryptography.
- b) Convert plaintext "How are you" into cyphertext using polyalphabetic cipher technique. Use key as Bye.
- c) Explain steganography with neat diagram.
- d) What is digital signature? Write its advantages and disadvantages.

Q.4. Attempt any THREE.

12

- a) Describe any two types of firewall with its advantages and disadvantages.
- b) Explain DMZ in detail with neat diagram.
- c) Describe basic components of Intrusion detection system with neat diagram.
- d) Differentiate between NIDS and HIDS.

